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LINUX FILESYSTEMS AND FILE MANAGEMENT

Directories and Files

Most Common Filesystem Directories

Let's now explore some of the most common filesystem directories:

Table 5.6: List of Main Directories and Their Purposes

DIRECTORY	PURPOSE
<code>/</code>	Primary directory of the entire filesystem hierarchy.
<code>/bin</code>	Essential executable programs to start and run the system. This directory is linked (shortcut) to <code>/usr/bin</code> .
<code>/boot</code>	Files required to boot the system. The kernel and support files live here.
<code>/dev</code>	Files used to represent the hardware devices installed on the system. Files are dynamically generated at boot time.
<code>/etc</code>	System-wide configuration files.
<code>/home</code>	User home directories, including personal settings, files, etc.
<code>/lib</code>	System library files. Kernel modules are stored in <code>/lib/modules</code> .
<code>/media</code>	Contains subdirectories used as mount points for removable media such as CDs, DVDs, USB sticks, etc.
<code>/mnt</code>	Mount point for temporarily mounted filesystems.
<code>/opt</code>	Suggested location for optional application software packages (third party applications).
<code>/proc</code>	Virtual pseudo-filesystem that provides information about the system and processes running on it. It is a dynamic memory-based directory. Can be used to alter system parameters.
<code>/run</code>	Runtime variable data that contains information about the system since it was booted (data associated with running processes). This directory is cleared at the start of the boot process.
<code>/sys</code>	Provides information (device, driver and some kernel) about the system and processes running on it. Can be used to alter system parameters.
<code>/root</code>	Home directory for the root user.
<code>/sbin</code>	Essential system binaries used by the root or other system administrators for booting, restoring, or repairing the operating system.
<code>/srv</code>	Data for services (HTTP and FTP) running on the server.
<code>/tmp</code>	Temporary files created by user applications.
<code>/usr</code>	Multi-user applications, utilities and data; theoretically read-only.

/var

Variable data that changes during system operation, including your system log files.